From Zero to Fast

Six Problems, Six Data Structures, One Music App

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Hi !

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THE IDEA

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RedisCover

a music streaming platform

a. **Daily Mixes** personalized playlists for each user
b. **Play Tracking** song play count tracking
c. **Top Songs** most-played songs
d. **Monthly Metrics** distinct listeners per artist
e. **Similar Songs** custom recommendations

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What are we going to do?

a. Clone a repo and spin up a sample app
b. Identify a problem
c. Measure it
d. Fix it with the right Redis data structure
e. Measure again to confirm the fix
f. Jump to b.

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`git clone`

https://github.com/paulorsousa/redis-music-workshop

~ `README.md` with all steps ~

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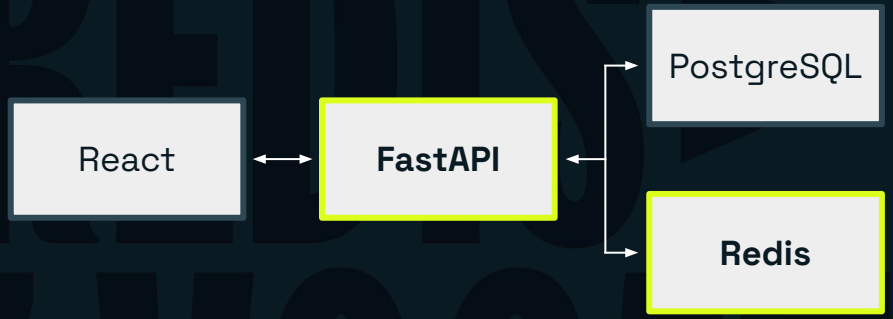
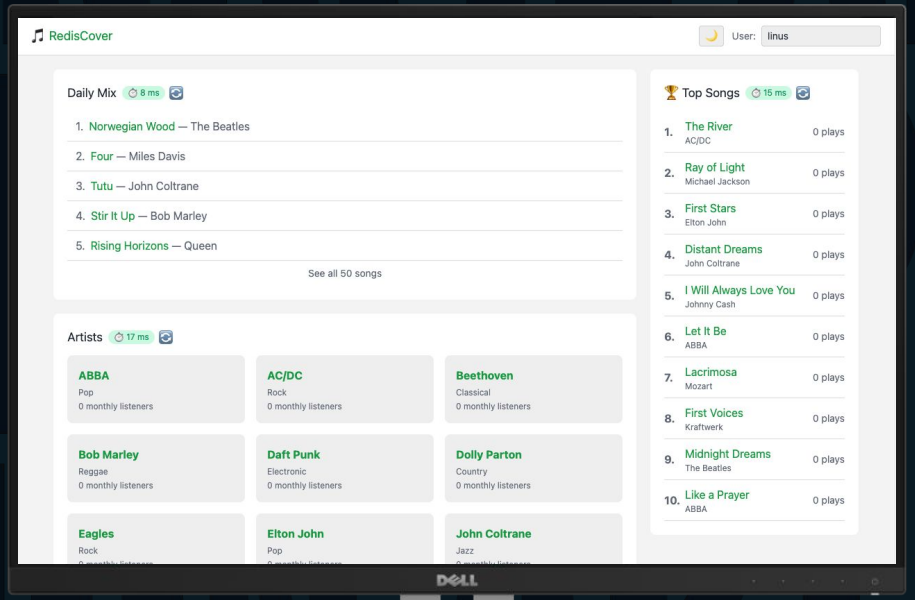
Architecture - Hello World!

React + Fast API + PostgreSQL + Redis





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1. docker compose up
2. ./workshop health
3. http://localhost:3000

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Module 1 – Caching

a. Daily mix algorithm is expensive

```
./workshop daily-mix
    (~5 seconds)
```

b. Runs every request

c. Use Redis as a cache

```
./workshop daily-mix
    (~10 milliseconds)
```

```
git checkout solution/module-1-caching
```

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Module 2 – Atomic counters

a. Concurrent play-count updates via SELECT + UPDATE cause lost writes in PostgreSQL

```
./workshop simulate-plays --song song-1
./workshop simulate-plays --song song-1--concurrent
      Actual:     8 ✗  (92 lost)
```

b. Check the UI: http://localhost:3000

c. Replace SQL calls with Redis INCR

```
./workshop reset
./workshop simulate-plays --song song-1--concurrent
      Actual:     100 ✓
```

```
git checkout solution/module-2-atomic-counters
```

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Module 3 – Sorted Sets

a. We just broke “Top Songs” leaderboard. Check the UI: http://localhost:3000

   Play-count is always zero, even after playing songs. Why?

b. Instead of a simple counter, we need to **keep songs sorted** by play-count

c. Redis Sorted Sets store elements ordered by score and support atomic score updates :)

d. Follow steps on README.md

```
git checkout solution/module-3-sorted-sets
```

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Module 4 – Sets

a. Besides every song play, RedisCover also counts each artist's unique monthly listeners

b. Current solution uses a list for each artist to store their monthly listeners

Problem?
⚠️ Inserting into an unsorted list while enforcing uniqueness is **O(N)** ⚠️

c. Replace the List by a Set (unordered)

d. Follow steps on README.md

```
git checkout solution/module-4-sets
```

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Module 5 – HyperLogLog

a. Redis Set solved the problem on performance, but…

b. Solution keeps using a lot of memory

   Each artist set stores the 36-char User IDs of each unique listener
   Our Top-5 artists have ~100M monthly listeners
   ⚠️ 36B * 100M * 5 ≈ **17 GB of memory per month** ⚠️

c. HyperLogLog is a probabilistic data structure that **estimates the cardinality of a set**.
   Trades perfect accuracy for efficient space utilization.
   Redis implementation uses up to 12 KB of memory for standard error rate of 0.81%

d. Follow steps on README.md

```
git checkout solution/module-5-hll
```

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Bonus: Module 6 – VectorSets

a. Redis 8.0 introduced a new data type: VectorSets
   Meaning we can now use core Redis as a in-memory **vector database**

b. What makes VectorSets exciting?

Memory-native & Fast	Built on Redis speed and efficiency
High-performance search	Approximate nearest neighbor (ANN) with HNSW (+info)
Rich metadata filtering	Filter by attributes, tags, or ranges
Scales with Redis	Cluster and replication support
Easy to use	Simple commands. No external vector DB needed

c. Great for real-world use cases

Recommendations	Recommended products, content, or users. Example: e-commerce “For you”
Fraud & Anomaly Detection	Find unusual patterns in behaviour or transactions. Example: flag suspicious logins
Code & Dev tools	Search code by natural language. Example: “Find function to parse JWT”
AI APPs & RAG Pipelines	Store embeddings and run queries against them. Example: chat with your docs

d. Follow steps on README.md

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QUESTIONS?

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